CyberShield OS

Security-Focused Mini Operating System

Laptop and System Requirements

1 Introduction

CyberShield OS is a lightweight, security-focused operating system project designed to demonstrate important Operating Systems concepts through system monitoring and security-related functionality. The project focuses on process monitoring, file integrity checking, login monitoring, system resource monitoring, and security alert generation.

Since the project does not involve developing a complete operating system kernel or performing computationally intensive machine learning tasks, it does not require high-end hardware. A typical student laptop is sufficient for development, testing, and demonstration.

2 Hardware Requirements

The following table describes the minimum and recommended laptop specifications for developing and running CyberShield OS.

Table 1: Laptop Hardware Requirements

Component	**Minimum Requirement**	**Recommended Requirement**
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5 / AMD Ryzen 5 or better
CPU Cores	4 cores	6 or more cores
RAM	8 GB	16 GB
Storage	256 GB SSD	512 GB SSD
Graphics	Integrated graphics	Integrated graphics is sufficient
Display	1366 × 768	1920 × 1080 (Full HD)
Internet	Required for software installation and updates	Stable Internet connection recommended

3 Software Requirements

The following software tools and platforms can be used for developing and testing CyberShield OS.

Table 2: Software Requirements

Software	**Recommended Platform / Version**
Operating System	Ubuntu 24.04 LTS or Kali Linux
Programming Language	Python 3.10 or later
Code Editor	Visual Studio Code or PyCharm
Version Control	Git and GitHub
Database	SQLite (Optional)
Virtualization Software	VMware Workstation or VirtualBox (Optional)
Terminal	Linux Terminal / Bash Shell

4 Development Environment

CyberShield OS can be developed directly on a Linux operating system or inside a virtual machine. Linux is preferred because the project interacts with system-level information such as processes, files, authentication logs, and resource usage.

A Linux environment also provides access to useful system interfaces such as the `/proc` filesystem and system log files.

The recommended development environment is:

- Ubuntu 24.04 LTS or Kali Linux.
- Python 3.10 or later.
- Visual Studio Code or another suitable code editor.
- Git and GitHub for collaborative development.
- Python virtual environment for dependency management.
- Terminal access for Linux commands and system monitoring.

Table 3: Virtual Machine Configuration

Resource	**Recommended Allocation**
Virtual RAM	4 GB
Virtual CPU	2 CPU cores
Virtual Disk	30–40 GB
Network Adapter	NAT or Bridged Adapter
Virtualization Technology	Intel VT-x / AMD-V enabled

5 Virtual Machine Requirements

If the project is developed or tested using VMware Workstation or VirtualBox, the following virtual machine configuration is recommended.

For laptops with 16 GB or more RAM, the virtual machine can be configured with 6–8 GB RAM and 4 CPU cores when additional resources are required.

It is recommended to avoid allocating excessive RAM or CPU resources to the virtual machine because the host operating system also requires sufficient resources.

6 Required Python Libraries

CyberShield OS can be implemented using Python and a small number of standard and third-party libraries.

Table 4: Python Libraries and Their Purpose

Library	**Purpose**
`psutil`	Process, CPU, memory, disk, and system monitoring
`hashlib`	File hashing and integrity verification
`os`	File system and operating system interaction
`subprocess`	Execution of Linux system commands
`datetime`	Timestamp and event management
`logging`	Security event and activity logging
`tkinter`	Optional graphical user interface
`sqlite3`	Optional storage of security events and alerts

7 System Modules and Resource Usage

The major modules of CyberShield OS are lightweight and can run on ordinary student laptops.

Table 5: CyberShield OS Modules

Module	**Function**
Process Monitor	Monitors currently running processes, process IDs, CPU usage, memory usage, and process information.
File Integrity Monitor	Uses cryptographic hashes to detect unauthorized or unexpected file modifications.
Login Monitor	Analyzes authentication logs and identifies repeated failed login attempts.
Resource Monitor	Monitors CPU, RAM, disk usage, and other system resources.
Security Alert System	Combines events from different modules and generates security alerts based on predefined rules.
Dashboard	Displays system status, security events, resource usage, and alerts in a user-friendly interface.

8 Recommended Laptop Configuration

For a smooth development and demonstration experience, the following configuration is recommended:

Component	**Recommended Specification**
Processor	Intel Core i5 / AMD Ryzen 5 or better
RAM	16 GB
Storage	512 GB SSD
Graphics	Integrated Graphics
Operating System	Windows 11 with Linux VM, or Linux directly
Virtual Machine RAM	4–8 GB
Virtual Machine CPU	2–4 cores

A dedicated graphics card is not required because CyberShield OS does not perform GPU-intensive computation. The most important resources for the project are CPU, RAM, and SSD storage.

9 Windows and VMware Configuration

Students using Windows 10 or Windows 11 can run CyberShield OS inside VMware Workstation or VirtualBox. For a laptop with 16 GB RAM, a suitable configuration is:

- Host Operating System: Windows 10/11.
- Virtual Machine Operating System: Ubuntu or Kali Linux.
- VM RAM: 4 GB initially.
- VM CPU: 2 cores initially.
- VM Storage: 30–40 GB.
- Network: NAT for normal Internet access or Bridged mode when required for network-related testing.
- Hardware virtualization: Intel VT-x or AMD-V enabled.

This configuration allows students to develop CyberShield OS while continuing to use normal Windows applications.

10 Performance Considerations

CyberShield OS is designed to have low resource consumption. The monitoring modules periodically collect information from the Linux system and process only the required data.

The project does not require:

- A dedicated graphics processing unit (GPU).
- 32 GB or more RAM.
- A high-end gaming laptop.
- Large-scale cloud infrastructure.
- High-performance workstation hardware.

An 8 GB RAM laptop can run the core project when CyberShield OS is executed directly on Linux. However, 16 GB RAM is recommended when Linux is executed inside a virtual machine alongside Windows.

Table 6: Minimum and Recommended System Configuration

Specification	**Minimum**	**Recommended**
CPU	Intel i3 / Ryzen 3	Intel i5 / Ryzen 5 or better
RAM	8 GB	16 GB
Storage	256 GB SSD	512 GB SSD
GPU	Integrated	Integrated
OS	Linux	Ubuntu/Kali Linux
VM RAM	2–4 GB	4–8 GB
VM CPU	2 cores	2–4 cores

11 Minimum vs Recommended Configuration

12 Conclusion

CyberShield OS has been designed to be suitable for a college-level Operating Systems project with a development period of approximately one month. The project focuses on practical system monitoring and security features without requiring high-end hardware.

The recommended configuration of an Intel Core i5 or AMD Ryzen 5 processor, 16 GB RAM, and a 512 GB SSD provides sufficient resources for development, testing, virtualization, and final demonstration.

For a five-member team, this configuration allows all members to develop and test their respective modules while maintaining a manageable and efficient development environment.

Recommended Configuration:

Intel Core i5 / Ryzen 5 + 16 GB RAM + 512 GB SSD